

# Evidence for New Sub-classification of Prime and Composite Numbers

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## Abstract

A new evidence is presented suggesting that prime and composite numbers can be further sub-classified into **additive** and **non-additive** sub-types in the decimal numbering system. Numbers **114** and **506** act as **twin numbers** to show the unmistakable evidence for this new number sub-classification.

**Keywords:** *number-theory, prime-numbers, composite-numbers, twin-numbers.*

## 1. Introduction

Prime and composite numbers play significant roles in cryptography as private and public keys respectively. Their decimal digit sums shed a new light on further sub-classification of these numbers into "additive" and "non-additive" sub-types based on the homogeneity between their types and the types of their digit sums.

The following sub-types have been uncovered by this work and are given different colours in the accompanying **PrimeCalculator** software for readability purposes:

- **BLUE** **Additive Prime:** Prime number with a prime digit sum.
- **GREEN** **Non-additive Prime:** Prime number with a non-prime digit sum.
- **RED** **Additive Composite:** Composite number with a composite digit sum.
- **BROWN** **Non-additive Composite:** Composite number with a non-composite digit sum.

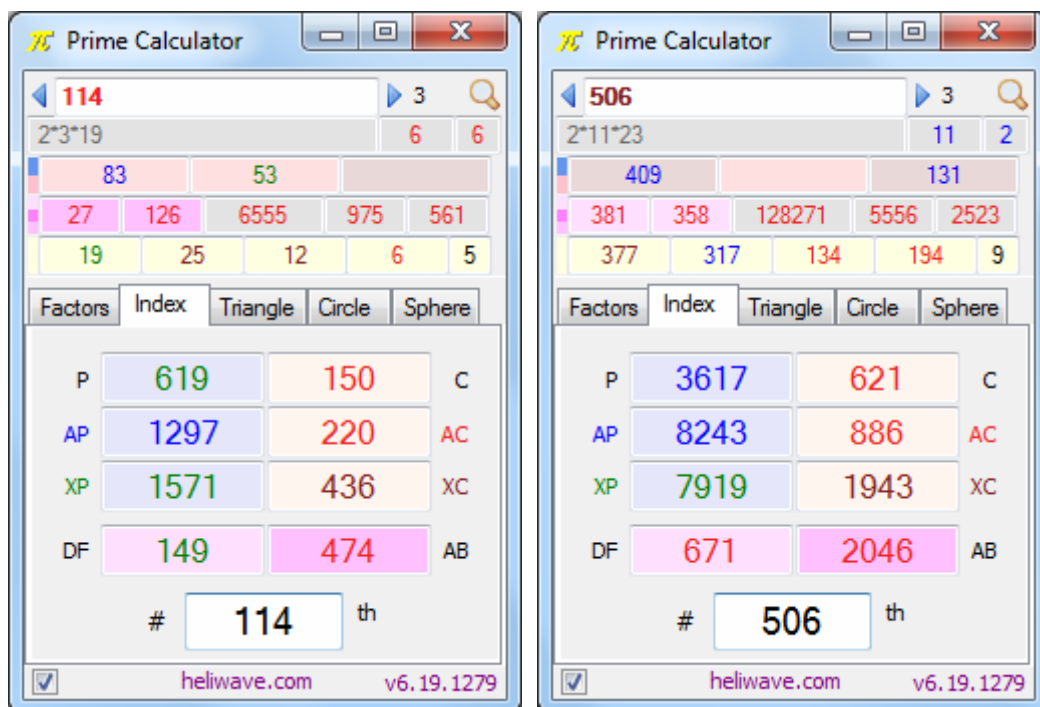


Fig 1: **PrimeCalculator** open-source software based on YAFU tool.  
Download from <http://heliwave.com/PrimeCalculator.zip>

## 2. Evidence for Number Sub-classification

The twin numbers **114** and **506** exhibit an unmistakable evidence for the need for new number sub-classification by sharing very similar number patterns and b) behaving as twin numbers when used as indices for prime and composite numbers.

### 2.1 Similar Number Patterns

Within the natural numbers, only **114** and **506** exhibit the following number pattern:

```
114 ÷ 2           = 57
median(1..57)    = 29
57 × 29          = 1653
The 16th prime    = 53
```

and,

```
506 ÷ 2           = 253
median(1..253)   = 127
253 × 127        = 32131
The 32nd prime   = 131
```

Furthermore,

There are exactly **16** prime numbers with prime digit sums up to **114**:  
**2, 3, 5, 7, 11, 23, 29, 41, 43, 47, 61, 67, 83, 89, 101, 113.**

There are exactly **53** composite numbers with composite digit sums up to **114**:  
**4, 6, 8, 9, 15, 18, 22, 24, 26, 27, 28, 33, 35, 36, 39, 40, 42, 44, 45, 46, 48, 51, 54, 55, 57, 60, 62, 63, 64, 66, 68, 69, 72, 75, 77, 78, 80, 81, 82, 84, 86, 87, 88, 90, 91, 93, 95, 96, 99, 105, 108, 112, 114.**

and,

There are exactly **32+10<sup>†</sup>** prime numbers with non-prime digit sums up to **506**:  
**13, 17, 19, 31, 37, 53, 59, 71, 73, 79, 97, 103, 107, 109, 127, 149, 163, 167, 181, 211, 233, 239, 251, 257, 271, 277, 293, 307, 347, 349, 367, 383, 389, 419, 431, 433, 439, 457, 479, 491, 499, 503.**

There are exactly **131** composite numbers with non-composite digit sums up to **506**:  
**10, 12, 14, 16, 20, 21, 25, 30, 32, 34, 38, 49, 50, 52, 56, 58, 65, 70, 74, 76, 85, 92, 94, 98, 100, 102, 104, 106, 110, 111, 115, 119, 120, 122, 124, 128, 133, 140, 142, 146, 148, 155, 160, 164, 166, 175, 182, 184, 188, 200, 201, 203, 205, 209, 210, 212, 214, 218, 221, 230, 232, 236, 238, 245, 247, 250, 254, 256, 265, 272, 274, 278, 287, 289, 290, 292, 296, 298, 300, 302, 304, 308, 319, 320, 322, 326, 328, 335, 340, 344, 346, 355, 362, 364, 368, 371, 377, 380, 382, 386, 388, 391, 395, 403, 407, 410, 412, 416, 418, 425, 427, 430, 434, 436, 445, 452, 454, 458, 469, 470, 472, 476, 478, 481, 485, 490, 494, 496, 500, 502, 506.**

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<sup>†</sup> **32+10**

32 non-additive primes with a digit sum != 10  
and  
10 non-additive primes with a digit sum == 10  
{ **19, 37, 73, 109, 127, 163, 181, 271, 307, 433** }.  
where  
**19 + 37 + 73 + 109 + 127 + 163 + 181 + 271 + 307 + 433 = 1720**  
and the sum of the first 31 prime numbers from **2** to **127** = **1720**

## 2.2 Twin-Numbers Behaviour

```

114th prime = 619
506th composite = 621
where
114 + 506 = 620

114th composite = 150
114th additive composite = 220
114th non-additive composite = 436
where 436 + 220 - 150 = 506

506th composite = 621
506th additive composite = 886
506th non-additive composite = 1943
where 1943 - 886 - 621 = 436
= 114th non-additive composite
and,

```

Sum of digit sums of numbers from 1 to 114 = 975

$$1+2+3+4+5+6+7+8+9+1+2+3+4+5+6+7+8+9+10+2+3+4+5+6+7+8+9+10+11+3+4+5+6+7+8+9+10+11+12+4+5+6+7+8+9+10+11+12+13+5+6+7+8+9+10+11+12+13+14+6+7+8+9+10+11+12+13+14+15+7+8+9+10+11+12+13+14+15+16+8+9+10+11+12+13+14+15+16+17+9+10+11+12+13+14+15+16+17+18+1+2+3+4+5+6+7+8+9+10+2+3+4+5+6$$

where

|                                      |                   |      |                                     |
|--------------------------------------|-------------------|------|-------------------------------------|
| 114 <sup>th</sup> prime              | =                 | 619  |                                     |
| 114 <sup>th</sup> additive prime     | =                 | 1297 |                                     |
| 114 <sup>th</sup> non-additive prime | =                 | 1571 |                                     |
|                                      | 619 + 1297 + 1571 | =    | 3487 = 975 <sup>th</sup> semi-prime |

and

[illegible]

Semi-prime index calculations: <http://heliwave.com/Composites.zip>

### 3. Further Work

The above interlocked relationships between the numbers **114** and **506** reveals the inherent sub-classing primes and composite numbers into additive and non-additive sub-types within the natural decimal numbers. An exhaustive computer search for natural numbers that exhibit the same pattern as the number **114** and **506**, namely *"n/2 multiplied by median(1...(n/2))" produces a triangular number as a concatenated prime and its index* yielded no other integer. However, using rational numbers, two triangular numbers were found with the same pattern:

$796 \div 2 = 398$   
 $\text{median}(1..398) = 199.5$   
 $398 \times 199.5 = 79401$   
 where  $79^{\text{th}}$  prime = 401

$37238756 \div 2 = 18619378$   
 $\text{median}(1..18619378) = 9309689.5$   
 $18619378 \times \underline{9309689.5} = 173340627863131$   
 where  $1733406^{\text{th}}$  prime = 27863131

For composites, only number **13444** was found to exhibit a similar pattern to the number 114 and 506 producing a concatenated composite (instead of prime) and its index:

**13444** ÷ 2 = 6722  
6722 ÷ 2 = 3361  
6722 × 3361 = 22592642

where

2259<sup>th</sup> composite = 2642

Further search for Index|Prime and Index|Composite returned no results.

Software tools used for verification:

<http://heliwave.com/IP.zip> and <http://heliwave.com/IC.zip>

#### 4. Conclusion

Since division by **1** does not divide a number into smaller parts, division by **1** should be treated as a trivial division (in the same sense as the trivial zeros of the zeta function) and can therefore be ignored. This will make the number **1** naturally indivisible without the need for the fixing of the definition of prime numbers to exclude **1** artificially. After all, **1** is *The Unit* (i.e. indivisible).

The following definitions arise naturally after excluding division by **1**:

|                        |      |                                                       |
|------------------------|------|-------------------------------------------------------|
| The Unit               | {U}  | = Whole number that is indivisible                    |
| Prime number           | {P}  | = Whole number that is divisible by itself only       |
| Additive Prime         | {AP} | = Prime with a prime digit sum                        |
| Non-additive Prime     | {XP} | = Prime with a non-prime digit sum                    |
| Composite number       | {C}  | = Whole number that is divisible by itself and others |
| Additive Composite     | {AC} | = Composite with a composite digit sum                |
| Non-additive Composite | {XC} | = Composite with a non-composite digit sum            |

The decimal system naturally lends itself to sub-classification of numbers based on their digit sums. If the type of a number and the type of its digit sum are the same, then it is called by this work "additive", else "non-additive" number.

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